

# RW4 Primary Chain Consolidated Summary

## INTERPRETATION GUIDE

- ADF  $p > 0.05$  and KPSS  $p < 0.05$  -> likely  $I(1)$ -like; prefer differences/cointegration models.
- ADF  $p < 0.05$  and KPSS  $p > 0.05$  -> stationary; level models are more admissible.
- Ljung-Box  $p < 0.05$  -> autocorrelation; add lag structure.
- BP/White  $p < 0.05$  -> heteroskedasticity; use robust or HAC standard errors.
- JB  $p < 0.05$  -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

## RUN NOTES

RW4 domestic vertical chain is FarmGateUA -> ProducerUA -> ProZorro -> Retail, estimated with both forward and reverse-flow pairs. Farm-gate enters from two alternative reconstruction workbooks and both linear/pchip interpolation variants are carried end-to-end.

panel\_count=104

model\_rows=2574

reverse\_rows=1137

benchmark\_rows=224

Table: Consolidated\_ModelCoefficients

| panel_level | panel_name                   | segment_name | product | standardized_type | retailer_panel | brand | price_variant | reconstruction_variant | farm_gate_source | frequency | chain_direction | stage_from | stage_to   | link                     | model_family        | y_series   | x_series   | n_obs | sr_coef                | lr_coef              | ect_coef             | ect_pvalue             | asymmetry_short_p   | asymmetry_long_p    | vecm_rank  | model_status | unreliable_flag           | unreliable_reason         | shock_dummy_included | promo_controls_included | ljungbox_p              | arch_p                       | jb_p                         | notes                        | robust_linear_vs_pchip | interpolation_sensitive | robust_across_reconstruction | core_finding_flag |   |   |  |
|-------------|------------------------------|--------------|---------|-------------------|----------------|-------|---------------|------------------------|------------------|-----------|-----------------|------------|------------|--------------------------|---------------------|------------|------------|-------|------------------------|----------------------|----------------------|------------------------|---------------------|---------------------|------------|--------------|---------------------------|---------------------------|----------------------|-------------------------|-------------------------|------------------------------|------------------------------|------------------------------|------------------------|-------------------------|------------------------------|-------------------|---|---|--|
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | forward         | FarmGateUA | ProducerUA | farmgateua_to_producerua | NO FIT              | ProducerUA | FarmGateUA | 41    | nan                    | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | unreliable   | 1                         | i2_series_blocked         | 0                    | 0                       | nan                     | nan                          | nan                          | i2_series_blocked            | 0                      | 0                       | 0                            |                   |   |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | forward         | ProducerUA | ProZorro   | producerua_to_prozorro   | NO FIT              | ProZorro   | ProducerUA | 41    | nan                    | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | unreliable   | 1                         | i2_series_blocked         | 0                    | 0                       | nan                     | nan                          | nan                          | i2_series_blocked            | 0                      | 0                       | 0                            |                   |   |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | forward         | ProZorro   | Retail     | prozorro_to_retail       | ARDL                | Retail     | ProZorro   | 41    | 0.03815343029021434    | -0.07780863784892769 | nan                  | nan                    | nan                 | nan                 | nan        | ok           | 0                         | 0                         | 0.8780631987617009   | 0.27069012089185107     | 0.65770037746761        | bounds_lower_p=0.0001846;... | 1                            | 0                            | 1                      | 0                       | 0                            |                   |   |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | forward         | ProZorro   | Retail     | prozorro_to_retail       | NARDL               | Retail     | ProZorro   | 39    | 0.1855195387021934     | 0.2943590807532827   | -0.7295217442383444  | 3.236682279562513e-10  | 0.06257081721067842 | 0.28322473471156157 | nan        | ok           | 0                         | 0                         | 0.8572002304658171   | 0.3274282228162803      | 0.6241106480263054      | Wald short/long asymmetry... | 1                            | 0                            | 1                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | forward         | ProZorro   | Retail     | prozorro_to_retail       | OLS_HAC_SHOCK       | Retail     | ProZorro   | 39    | 0.05139989377987775    | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | ok           | 0                         | 0                         | 0.8363278130131624   | 0.8637032834700069      | 0.5993001807630483      | shock_coef=0.02914; shock... | 1                            | 0                            | 1                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | forward         | ProZorro   | Retail     | prozorro_to_retail       | OLS_HAC_RETAIL_CTRL | Retail     | ProZorro   | 39    | 0.05444419625534165    | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | ok           | 0                         | 0                         | 0.7635580864009127   | 0.7176527168689669      | 0.7798013280299441      | controls=dy_l1,dx,shock...   | 1                            | 0                            | 1                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | forward         | FarmGateUA | ProZorro   | farmgateua_to_prozorro   | ARDL                | ProZorro   | FarmGateUA | 41    | 27.40581164655441      | -1.5633672197814128  | nan                  | nan                    | nan                 | nan                 | unreliable | 1            | residual_autocorr_or_arch | 0                         | 0                    | 0.04265890648236325     | 0.947046008010218       | 0.1379433157208637           | bounds_lower_p=0.5287; bo... | 1                            | 0                      | 1                       | 0                            | 0                 | 0 |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | forward         | FarmGateUA | ProZorro   | farmgateua_to_prozorro   | NARDL               | ProZorro   | FarmGateUA | 39    | -39.44395541420697     | -18.226599084282483  | -1.3944276767203982  | 9.480407060354374e-12  | 0.32320473925984067 | 0.09472115674979945 | nan        | unreliable   | 1                         | residual_autocorr_or_arch | 0                    | 0                       | 0.01792346097271439     | 0.9900728576475119           | 0.0012442109522715625        | Wald short/long asymmetry... | 1                      | 0                       | 1                            | 0                 | 0 | 0 |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | forward         | FarmGateUA | ProZorro   | farmgateua_to_prozorro   | OLS_HAC_SHOCK       | ProZorro   | FarmGateUA | 39    | -7.168698194945932     | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | unreliable   | 1                         | residual_autocorr_or_arch | 0                    | 0                       | 0.04793189025212023     | 0.364387368464964            | 0.5147495938242576           | shock_coef=-0.04466; shoc... | 1                      | 0                       | 1                            | 0                 | 0 | 0 |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | forward         | ProducerUA | Retail     | producerua_to_retail     | NO FIT              | Retail     | ProducerUA | 41    | nan                    | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | unreliable   | 1                         | i2_series_blocked         | 0                    | 0                       | nan                     | nan                          | nan                          | i2_series_blocked            | 0                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | forward         | FarmGateUA | Retail     | farmgateua_to_retail     | NARDL               | Retail     | FarmGateUA | 39    | -8.772023103981054     | -1.852043589957451   | -0.7869901901005358  | 1.6544457942157222e-06 | 0.5313232286712996  | 0.6001897201575894  | nan        | ok           | 0                         | 0                         | 0.6150515057911954   | 0.0893311175436191      | 0.3721932861070034      | Wald short/long asymmetry... | 1                            | 0                            | 1                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | forward         | FarmGateUA | Retail     | farmgateua_to_retail     | OLS_HAC_SHOCK       | Retail     | FarmGateUA | 39    | -5.584451677878315     | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | ok           | 0                         | 0                         | 0.8407893391041152   | 0.6042533997790516      | 0.7048762421900949      | shock_coef=0.03516; shock... | 1                            | 0                            | 1                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | forward         | FarmGateUA | Retail     | farmgateua_to_retail     | OLS_HAC_RETAIL_CTRL | Retail     | FarmGateUA | 39    | -7.4832051162984845    | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | ok           | 0                         | 0                         | 0.7212890938109395   | 0.9821125911073111      | 0.6896754473970916      | controls=dy_l1,dx,shock...   | 1                            | 0                            | 1                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | reverse         | Retail     | ProZorro   | retail_to_prozorro       | NARDL               | ProZorro   | Retail     | 39    | 1.562917141304475      | -0.2420696049936481  | -1.1193054614387603  | 6.869385961022385e-15  | 0.08959554902277322 | 0.72059350414332    | nan        | ok           | 0                         | 0                         | 0.5944183073130898   | 0.9317839276243874      | 2.9427177490413163e-09  | Wald short/long asymmetry... | 1                            | 0                            | 1                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | reverse         | Retail     | ProZorro   | retail_to_prozorro       | OLS_HAC_SHOCK       | ProZorro   | Retail     | 39    | 0.738873459637164      | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | ok           | 0                         | 0                         | 0.05437647925022223  | 0.44781824112115143     | 0.7019279381875408      | shock_coef=-0.07536; shoc... | 1                            | 0                            | 1                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | reverse         | ProZorro   | ProducerUA | prozorro_to_producerua   | NO FIT              | ProducerUA | ProZorro   | 41    | nan                    | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | unreliable   | 1                         | i2_series_blocked         | 0                    | 0                       | nan                     | nan                          | nan                          | i2_series_blocked            | 0                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | reverse         | Retail     | ProducerUA | retail_to_producerua     | NO FIT              | ProducerUA | Retail     | 41    | nan                    | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | unreliable   | 1                         | i2_series_blocked         | 0                    | 0                       | nan                     | nan                          | nan                          | i2_series_blocked            | 0                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | reverse         | ProducerUA | FarmGateUA | producerua_to_farmgateua | NO FIT              | FarmGateUA | ProducerUA | 41    | nan                    | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | unreliable   | 1                         | i2_series_blocked         | 0                    | 0                       | nan                     | nan                          | nan                          | i2_series_blocked            | 0                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | reverse         | ProZorro   | FarmGateUA | prozorro_to_farmgateua   | NARDL               | FarmGateUA | ProZorro   | 39    | 0.0012943707500721905  | 0.021025715379738956 | -0.13491630656349962 | 0.002529451476566982   | 0.723653936801244   | 0.6356065349486733  | nan        | unreliable   | 1                         | residual_autocorr_or_arch | 0                    | 0                       | 0.2694081554124808      | 0.005653264141954011         | 0.7148836239452375           | Wald short/long asymmetry... | 1                      | 0                       | 1                            | 0                 | 0 | 0 |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | reverse         | ProZorro   | FarmGateUA | prozorro_to_farmgateua   | OLS_HAC_SHOCK       | FarmGateUA | ProZorro   | 39    | 0.0005263704198728194  | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | unreliable   | 1                         | residual_autocorr_or_arch | 0                    | 0                       | 0.5771314315458762      | 0.000760141130922262         | 1.2496837180416046e-15       | shock_coef=0.0007883; sho... | 1                      | 0                       | 1                            | 0                 | 0 | 0 |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | reverse         | Retail     | FarmGateUA | retail_to_farmgateua     | NARDL               | FarmGateUA | Retail     | 39    | 0.006744790717442998   | 0.0979596312980813   | -0.14487231326836603 | 9.1537068688685441e-05 | 0.44380481051615583 | 0.29882002822034315 | nan        | unreliable   | 1                         | residual_autocorr_or_arch | 0                    | 0                       | 0.5866962264906624      | 0.025510363550298523         | 0.42003209957856735          | Wald short/long asymmetry... | 1                      | 0                       | 1                            | 0                 | 0 | 0 |  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки  | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | reverse         | Retail     | FarmGateUA | retail_to_farmgateua     | OLS_HAC_SHOCK       | FarmGateUA | Retail     | 39    | -0.0027240845359216854 | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | unreliable   | 1                         | residual_autocorr_or_arch | 0                    | 0                       | 0.714082605152659       | 0.0007160424494383612        | 7.857516478631708e-14        | shock_coef=0.00086; shock... | 1                      | 0                       | 1                            | 0                 | 0 | 0 |  |
| product     | product::Молоко питне::Si... | Молоко       | питне   | Молоко            | питне          | Silpo | observed      | linear                 | initial          | daily     | forward         | FarmGateUA | ProducerUA | farmgateua_to_producerua | NO FIT              | ProducerUA | FarmGateUA | 41    | nan                    | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | unreliable   | 1                         | i2_series_blocked         | 0                    | 0                       | nan                     | nan                          | nan                          | i2_series_blocked            | 0                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Молоко питне::Si... | Молоко       | питне   | Молоко            | питне          | Silpo | observed      | linear                 | initial          | daily     | forward         | ProducerUA | ProZorro   | producerua_to_prozorro   | NARDL               | ProZorro   | ProducerUA | 39    | -39.35627563072685     | -1.428284989733576   | -0.9847510668431191  | 6.743108432211024e-17  | 0.36545679796484387 | 0.8822095970812577  | nan        | ok           | 0                         | 0                         | 0.9709884811190265   | 0.9999753593423056      | 1.1527164906731183e-235 | Wald short/long asymmetry... | 1                            | 0                            | 1                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Молоко питне::Si... | Молоко       | питне   | Молоко            | питне          | Silpo | observed      | linear                 | initial          | daily     | forward         | ProducerUA | ProZorro   | producerua_to_prozorro   | OLS_HAC_SHOCK       | ProZorro   | ProducerUA | 39    | 1.19463804443748004    | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | ok           | 0                         | 0                         | 0.5747205516915184   | 0.6082698942466701      | 7.141746996589135e-42   | shock_coef=-0.01855; shoc... | 1                            | 0                            | 1                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Молоко питне::Si... | Молоко       | питне   | Молоко            | питне          | Silpo | observed      | linear                 | initial          | daily     | forward         | ProZorro   | Retail     | prozorro_to_retail       | NARDL               | Retail     | ProZorro   | 39    | -0.051509672665930616  | -0.04955189374108957 | -0.5842457751880188  | 0.0002182412990053074  | 0.5149646213320693  | 0.8578182729176604  | nan        | ok           | 0                         | 0                         | 0.9270232156217034   | 0.25425800530834386     | 0.36083890651554734     | Wald short/long asymmetry... | 1                            | 0                            | 1                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Молоко питне::Si... | Молоко       | питне   | Молоко            | питне          | Silpo | observed      | linear                 | initial          | daily     | forward         | ProZorro   | Retail     | prozorro_to_retail       | OLS_HAC_SHOCK       | Retail     | ProZorro   | 39    | -0.03448134071127669   | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | ok           | 0                         | 0                         | 0.641300532061247    | 0.21392336243510822     | 0.8212541816502396      | shock_coef=0.001439; shoc... | 1                            | 0                            | 1                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Молоко питне::Si... | Молоко       | питне   | Молоко            | питне          | Silpo | observed      | linear                 | initial          | daily     | forward         | ProZorro   | Retail     | prozorro_to_retail       | OLS_HAC_RETAIL_CTRL | Retail     | ProZorro   | 39    | -0.02323374756997762   | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | ok           | 0                         | 0                         | 0.3125667229530966   | 0.8338347694488211      | 0.8680000712778161      | controls=dy_l1,dx,shock...   | 1                            | 0                            | 1                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Молоко питне::Si... | Молоко       | питне   | Молоко            | питне          | Silpo | observed      | linear                 | initial          | daily     | forward         | FarmGateUA | ProZorro   | farmgateua_to_prozorro   | NO FIT              | ProZorro   | FarmGateUA | 41    | nan                    | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | unreliable   | 1                         | i2_series_blocked         | 0                    | 0                       | nan                     | nan                          | nan                          | i2_series_blocked            | 0                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Молоко питне::Si... | Молоко       | питне   | Молоко            | питне          | Silpo | observed      | linear                 | initial          | daily     | forward         | ProducerUA | Retail     | producerua_to_retail     | NARDL               | Retail     | ProducerUA | 39    | -15.298524573152017    | 1.1437855475284981   | -0.7215120205461834  | 9.127741123168788e-07  | 0.2902907120428701  | 0.7245139755845282  | nan        | ok           | 0                         | 0                         | 0.9804362238233978   | 0.16188209007499407     | 0.03021573381316071     | Wald short/long asymmetry... | 1                            | 0                            | 1                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Молоко питне::Si... | Молоко       | питне   | Молоко            | питне          | Silpo | observed      | linear                 | initial          | daily     | forward         | ProducerUA | Retail     | producerua_to_retail     | OLS_HAC_SHOCK       | Retail     | ProducerUA | 39    | -0.7555307082083995    | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | ok           | 0                         | 0                         | 0.9084146038366925   | 0.8278286357625326      | 0.8378889744553367      | shock_coef=-0.0002129; sh... | 1                            | 0                            | 1                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Молоко питне::Si... | Молоко       | питне   | Молоко            | питне          | Silpo | observed      | linear                 | initial          | daily     | forward         | ProducerUA | Retail     | producerua_to_retail     | OLS_HAC_RETAIL_CTRL | Retail     | ProducerUA | 39    | -0.4191986336309853    | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | ok           | 0                         | 0                         | 1                    | 0                       | 0.416733110421963       | 0.9341571860320718           | 0.7033144911317335           | controls=dy_l1,dx,shock...   | 1                      | 0                       | 1                            | 0                 | 0 | 0 |  |
| product     | product::Молоко питне::Si... | Молоко       | питне   | Молоко            | питне          | Silpo | observed      | linear                 | initial          | daily     | forward         | FarmGateUA | Retail     | farmgateua_to_retail     | NO FIT              | Retail     | FarmGateUA | 41    | nan                    | nan                  | nan                  | nan                    | nan                 | nan                 | nan        | unreliable   | 1                         | i2_series_blocked         | 0                    | 0                       | nan                     | nan                          | nan                          | i2_series_blocked            | 0                      | 0                       | 0                            | 0                 |   |   |  |
| product     | product::Молоко питне::Si... | Молоко       | питне   | Молоко            | питне          | Silpo | observed      | linear                 | initial          | daily     | reverse         | Retail     | ProZorro   | retail_to_prozorro       | ARDL                | ProZorro   | Retail     | 41    | -1.569298877962051     | 1.4624940242583346   | nan                  | nan                    | nan                 | nan                 | nan        | ok           | 0                         | 0                         | 0.8435337154070991   | 0.9672870409827264      | 1.494891740049852e-     |                              |                              |                              |                        |                         |                              |                   |   |   |  |

Table: Variant\_Robustness

| panel_level | segment_name         | standardized_type    | retailer_panel | brand | price_variant | farm_gate_source   | chain_direction | stage_from | stage_to   | model_family | same_sr_sign | same_lr_sign | same_ect_sign | robust_linear_vs_pchip | interpolation_sensitive |
|-------------|----------------------|----------------------|----------------|-------|---------------|--------------------|-----------------|------------|------------|--------------|--------------|--------------|---------------|------------------------|-------------------------|
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | forward         | FarmGateUA | ProZorro   | NARDL        | True         | True         | True          | 1                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | forward         | FarmGateUA | ProZorro   | VECM         | False        | False        | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | forward         | FarmGateUA | ProducerUA | ARDL         | True         | False        | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | forward         | FarmGateUA | ProducerUA | NARDL        | True         | False        | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | forward         | FarmGateUA | Retail     | NARDL        | True         | True         | True          | 1                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | forward         | FarmGateUA | Retail     | VECM         | False        | False        | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | forward         | ProZorro   | Retail     | NARDL        | True         | True         | True          | 1                      | 0                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | forward         | ProducerUA | ProZorro   | NARDL        | True         | True         | True          | 1                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | forward         | ProducerUA | Retail     | ARDL         | nan          | nan          | nan           | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | forward         | ProducerUA | Retail     | NARDL        | True         | True         | True          | 1                      | 0                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | forward         | ProducerUA | Retail     | VECM         | False        | False        | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | reverse         | ProZorro   | FarmGateUA | NARDL        | False        | False        | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | reverse         | ProZorro   | ProducerUA | NARDL        | False        | False        | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | reverse         | ProducerUA | FarmGateUA | ARDL         | True         | True         | True          | 1                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | reverse         | ProducerUA | FarmGateUA | NARDL        | True         | True         | True          | 1                      | 0                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | reverse         | Retail     | FarmGateUA | ARDL         | False        | True         | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | reverse         | Retail     | FarmGateUA | NARDL        | True         | True         | True          | 1                      | 0                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | reverse         | Retail     | ProZorro   | NARDL        | True         | True         | True          | 1                      | 0                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | reverse         | Retail     | ProducerUA | ARDL         | True         | True         | True          | 1                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | all_missing_filled | reverse         | Retail     | ProducerUA | NARDL        | True         | True         | True          | 1                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | forward         | FarmGateUA | ProZorro   | NARDL        | True         | True         | True          | 1                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | forward         | FarmGateUA | ProZorro   | VECM         | False        | False        | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | forward         | FarmGateUA | ProducerUA | ARDL         | nan          | nan          | nan           | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | forward         | FarmGateUA | ProducerUA | NARDL        | False        | True         | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | forward         | FarmGateUA | Retail     | NARDL        | True         | True         | True          | 1                      | 0                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | forward         | FarmGateUA | Retail     | VECM         | False        | False        | False         | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | forward         | ProZorro   | Retail     | NARDL        | True         | True         | True          | 1                      | 0                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | forward         | ProducerUA | ProZorro   | NARDL        | True         | True         | True          | 1                      | 0                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | forward         | ProducerUA | Retail     | ARDL         | nan          | nan          | nan           | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | forward         | ProducerUA | Retail     | NARDL        | True         | True         | True          | 1                      | 0                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | forward         | ProducerUA | Retail     | VECM         | False        | False        | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | reverse         | ProZorro   | FarmGateUA | ARDL         | True         | True         | True          | 1                      | 0                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | reverse         | ProZorro   | FarmGateUA | NARDL        | False        | False        | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | reverse         | ProZorro   | ProducerUA | NARDL        | False        | False        | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | reverse         | ProducerUA | FarmGateUA | ARDL         | True         | False        | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | reverse         | ProducerUA | FarmGateUA | NARDL        | False        | True         | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | reverse         | Retail     | FarmGateUA | ARDL         | False        | True         | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | reverse         | Retail     | FarmGateUA | NARDL        | False        | False        | True          | 0                      | 1                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | reverse         | Retail     | ProZorro   | NARDL        | True         | True         | True          | 1                      | 0                       |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | initial            | reverse         | Retail     | ProducerUA | ARDL         | True         | True         | True          | 1                      | 0                       |

Table: FarmGate\_Source\_Comparison

| panel_level | segment_name         | standardized_type    | retailer_panel | brand | price_variant | reconstruction_variant | chain_direction | stage_from | stage_to   | model_family | robust_across_reconstruction |
|-------------|----------------------|----------------------|----------------|-------|---------------|------------------------|-----------------|------------|------------|--------------|------------------------------|
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | forward         | FarmGateUA | ProZorro   | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | forward         | FarmGateUA | ProZorro   | VECM         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | forward         | FarmGateUA | ProducerUA | ARDL         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | forward         | FarmGateUA | ProducerUA | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | forward         | FarmGateUA | Retail     | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | forward         | FarmGateUA | Retail     | VECM         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | forward         | ProZorro   | Retail     | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | forward         | ProducerUA | ProZorro   | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | forward         | ProducerUA | Retail     | ARDL         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | forward         | ProducerUA | Retail     | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | forward         | ProducerUA | Retail     | VECM         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | reverse         | ProZorro   | FarmGateUA | ARDL         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | reverse         | ProZorro   | FarmGateUA | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | reverse         | ProZorro   | ProducerUA | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | reverse         | ProducerUA | FarmGateUA | ARDL         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | reverse         | ProducerUA | FarmGateUA | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | reverse         | Retail     | FarmGateUA | ARDL         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | reverse         | Retail     | FarmGateUA | NARDL        | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | reverse         | Retail     | ProZorro   | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | reverse         | Retail     | ProducerUA | ARDL         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | linear                 | reverse         | Retail     | ProducerUA | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | forward         | FarmGateUA | ProZorro   | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | forward         | FarmGateUA | ProZorro   | VECM         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | forward         | FarmGateUA | ProducerUA | ARDL         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | forward         | FarmGateUA | ProducerUA | NARDL        | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | forward         | FarmGateUA | Retail     | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | forward         | FarmGateUA | Retail     | VECM         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | forward         | ProZorro   | Retail     | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | forward         | ProducerUA | ProZorro   | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | forward         | ProducerUA | Retail     | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | forward         | ProducerUA | Retail     | VECM         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | reverse         | ProZorro   | FarmGateUA | ARDL         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | reverse         | ProZorro   | FarmGateUA | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | reverse         | ProZorro   | ProducerUA | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | reverse         | ProducerUA | FarmGateUA | ARDL         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | reverse         | ProducerUA | FarmGateUA | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | reverse         | Retail     | FarmGateUA | ARDL         | 0                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | reverse         | Retail     | FarmGateUA | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | reverse         | Retail     | ProZorro   | NARDL        | 1                            |
| average     | all_products_average | all_products_average | Silpo          |       | baseline      | pchip                  | reverse         | Retail     | ProducerUA | ARDL         | 0                            |

Table: Benchmark\_Comparison

| panel_level | panel_name                   | segment_name | product      | standardized_type | retailer_panel | brand | price_variant | reconstruction_variant | farm_gate_source | frequency | benchmark_source | stage     | n_obs | best_lag_days | corr_at_best_lag     |
|-------------|------------------------------|--------------|--------------|-------------------|----------------|-------|---------------|------------------------|------------------|-----------|------------------|-----------|-------|---------------|----------------------|
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки       | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | EU               | farm_gate | 35    | 8             | -0.5321612786428109  |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки       | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | EU               | producer  | 35    | 13            | -0.440450247374384   |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки       | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | EU               | prozorro  | 35    | 1             | -0.36658588531900926 |
| product     | product::Вершки::Silpo::o... | Вершки       | Вершки       | cream             | Silpo          |       | observed      | linear                 | initial          | daily     | EU               | retail    | 35    | 5             | -0.25023426876074867 |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | observed      | linear                 | initial          | daily     | ConsumerUA       | farm_gate | 41    | 14            | -0.35136269514440793 |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | observed      | linear                 | initial          | daily     | ConsumerUA       | producer  | 41    | 1             | 0.721980024121582    |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | observed      | linear                 | initial          | daily     | ConsumerUA       | prozorro  | 41    | 6             | -0.05060394142860878 |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | observed      | linear                 | initial          | daily     | ConsumerUA       | retail    | 41    | 3             | 0.1769670390737025   |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | observed      | linear                 | initial          | daily     | EU               | farm_gate | 35    | 2             | 0.5839116609801674   |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | observed      | linear                 | initial          | daily     | EU               | producer  | 35    | 13            | -0.4264206773893265  |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | observed      | linear                 | initial          | daily     | EU               | prozorro  | 35    | 13            | 0.28115777668820313  |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | observed      | linear                 | initial          | daily     | EU               | retail    | 35    | 1             | -0.3523422763099678  |
| product     | product::Сир твердый::Sil... | Сир твердый  | Сир твердый  | hard_cheese       | Silpo          |       | observed      | linear                 | initial          | daily     | ConsumerUA       | farm_gate | 38    | 14            | -0.4793962514673188  |
| product     | product::Сир твердый::Sil... | Сир твердый  | Сир твердый  | hard_cheese       | Silpo          |       | observed      | linear                 | initial          | daily     | ConsumerUA       | producer  | 38    | 1             | 0.4709771745526187   |
| product     | product::Сир твердый::Sil... | Сир твердый  | Сир твердый  | hard_cheese       | Silpo          |       | observed      | linear                 | initial          | daily     | ConsumerUA       | prozorro  | 38    | 8             | 0.26164643966008283  |
| product     | product::Сир твердый::Sil... | Сир твердый  | Сир твердый  | hard_cheese       | Silpo          |       | observed      | linear                 | initial          | daily     | ConsumerUA       | retail    | 38    | 13            | 0.21044538897242854  |
| product     | product::Сир твердый::Sil... | Сир твердый  | Сир твердый  | hard_cheese       | Silpo          |       | observed      | linear                 | initial          | daily     | EU               | farm_gate | 35    | 13            | 0.28300900383974525  |
| product     | product::Сир твердый::Sil... | Сир твердый  | Сир твердый  | hard_cheese       | Silpo          |       | observed      | linear                 | initial          | daily     | EU               | producer  | 35    | 13            | -0.709133777245297   |
| product     | product::Сир твердый::Sil... | Сир твердый  | Сир твердый  | hard_cheese       | Silpo          |       | observed      | linear                 | initial          | daily     | EU               | prozorro  | 35    | 8             | -0.3167871985828982  |
| product     | product::Сир твердый::Sil... | Сир твердый  | Сир твердый  | hard_cheese       | Silpo          |       | observed      | linear                 | initial          | daily     | EU               | retail    | 35    | 10            | -0.26659294614669893 |
| product     | product::Сметана::Silpo::... | Сметана      | Сметана      | sour_cream        | Silpo          |       | observed      | linear                 | initial          | daily     | ConsumerUA       | farm_gate | 38    | 14            | -0.4207947300982381  |
| product     | product::Сметана::Silpo::... | Сметана      | Сметана      | sour_cream        | Silpo          |       | observed      | linear                 | initial          | daily     | ConsumerUA       | producer  | 38    | 1             | 0.7182857764710593   |
| product     | product::Сметана::Silpo::... | Сметана      | Сметана      | sour_cream        | Silpo          |       | observed      | linear                 | initial          | daily     | ConsumerUA       | prozorro  | 38    | 4             | -0.1604594825928894  |
| product     | product::Сметана::Silpo::... | Сметана      | Сметана      | sour_cream        | Silpo          |       | observed      | linear                 | initial          | daily     | ConsumerUA       | retail    | 38    | 13            | 0.2697208823590196   |
| product     | product::Вершки::Silpo::b... | Вершки       | Вершки       | cream             | Silpo          |       | baseline      | linear                 | initial          | daily     | EU               | farm_gate | 35    | 8             | -0.5321612786428109  |
| product     | product::Вершки::Silpo::b... | Вершки       | Вершки       | cream             | Silpo          |       | baseline      | linear                 | initial          | daily     | EU               | producer  | 35    | 13            | -0.440450247374384   |
| product     | product::Вершки::Silpo::b... | Вершки       | Вершки       | cream             | Silpo          |       | baseline      | linear                 | initial          | daily     | EU               | prozorro  | 35    | 1             | -0.36658588531900926 |
| product     | product::Вершки::Silpo::b... | Вершки       | Вершки       | cream             | Silpo          |       | baseline      | linear                 | initial          | daily     | EU               | retail    | 35    | 14            | 0.33720770407033374  |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | baseline      | linear                 | initial          | daily     | ConsumerUA       | farm_gate | 41    | 14            | -0.35136269514440793 |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | baseline      | linear                 | initial          | daily     | ConsumerUA       | producer  | 41    | 1             | 0.721980024121582    |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | baseline      | linear                 | initial          | daily     | ConsumerUA       | prozorro  | 41    | 6             | -0.05060394142860878 |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | baseline      | linear                 | initial          | daily     | ConsumerUA       | retail    | 41    | 12            | -0.2034854549625866  |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | baseline      | linear                 | initial          | daily     | EU               | farm_gate | 35    | 2             | 0.5839116609801674   |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | baseline      | linear                 | initial          | daily     | EU               | producer  | 35    | 13            | -0.4264206773893265  |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | baseline      | linear                 | initial          | daily     | EU               | prozorro  | 35    | 13            | 0.28115777668820313  |
| product     | product::Молоко питне::Si... | Молоко питне | Молоко питне | milk              | Silpo          |       | baseline      | linear                 | initial          | daily     | EU               | retail    | 35    | 1             | -0.31573471579523754 |
| product     | product::Сир твердый::Sil... | Сир твердый  | Сир твердый  | hard_cheese       | Silpo          |       | baseline      | linear                 | initial          | daily     | ConsumerUA       | farm_gate | 38    | 14            | -0.4793962514673188  |
| product     | product::Сир твердый::Sil... | Сир твердый  | Сир твердый  | hard_cheese       | Silpo          |       | baseline      | linear                 | initial          | daily     | ConsumerUA       | producer  | 38    | 1             | 0.4709771745526187   |
| product     | product::Сир твердый::Sil... | Сир твердый  | Сир твердый  | hard_cheese       | Silpo          |       | baseline      | linear                 | initial          | daily     | ConsumerUA       | prozorro  | 38    | 8             | 0.26164643966008283  |
| product     | product::Сир твердый::Sil... | Сир твердый  | Сир твердый  | hard_cheese       | Silpo          |       | baseline      | linear                 | initial          | daily     | ConsumerUA       | retail    | 38    | 12            | -0.21485071003446954 |